

Assignment#1

Q1. Identify three novel applications of the Internet or multimedia applications

Here are 3 novel applications of the internet or multimedia technologies

1. Telemedicine & Remote Surgery (Internet + Multimedia + AI)

Doctors can now diagnose and even treat patients remotely using real-time video, sensors, and connected medical devices

- A patient in a rural area connected with a specialist via high-definition video
- Medical data (heart rate, scans, etc) is streamed live over the internet
- In advanced cases, robotics surgical system can be controlled remotely

Why it's novel: It turns the internet into a real-time healthcare delivery system, not just Communication

Example:

- Babyl Rwanda: is one of the most well-known telemedicine system in the Country.

2. Immersive Education (AR/VR + Internet Learning Platform)

Education is evolving into interactive 3D environments using Augmented Reality(AR) and Virtual Reality(VR)

- Students can enter a virtual biology lab to explore the human body
- History classes can simulate ancient cities in 3D

Why it's novel: It replaces passive learning(reading/watching) with experiential learning Environment

Example:

- Zipline Rwanda: operates autonomous drones delivery medical supplies

3. Smart cities & IoT Multimedia Dashboards

Cities use connected sensors + internet systems to manage infrastructure in real time.

- Traffic cameras + sensor feed live data to AI systems
- Citizens see real-time dashboards (traffic, pollution, energy usage)
- Emergency services respond faster using live multimedia feeds

Why it's novel: The internet becomes the central nervous system of a city

Example:

- Kigali Smart city initiatives

Kigali is implementing smart infrastructure systems

- CCTV surveillance + traffic monitoring system
- Digital services for government (online applications, payments, reporting issues)

Q2: Discuss why adobe photoshop is commonly today as the photoshop editing software

Adobe photoshop is widely used today as a photo editing software because it combines power, flexibility, industry adoption, and continuous innovation in one ecosystem

1. Extremely Powerful Editing Capability

Adobe photoshop offers a level of control that most other tools don't match

- Advanced layer-based editing
- Precise color correction and retouching
- Complex selections(hair, objects, background)

2. Industry Standard in Creative Fields

Photoshop has become the default tool in design, photography, advertising, and media.

- Used by photographers, graphic designers, marketers and UI designer
- Most tutorials, courses, and job requirements are based on photoshop
- File formats (like PSD) are widely supported across tools

3. Strong Ecosystem & Integration

Adobe Photoshop is part of the broader Adobe ecosystem:

- Works seamlessly with illustrator, Lightroom, After Effects
- Cloud storage and collaboration via Adobe Creative Cloud
- Huge plugin marketplace (filters, AI tools, automation scripts)

4. AI and Modern Features (Recent Advantage)

Modern photoshop includes AI-powered tools

- Generative Fill (add/remove objects intelligently)
- Neural Filters (age change, style transfer, skin smoothing)
- Auto selection and background removal

5. Large Learning Resources & Community

- Millions of tutorials on YouTube, Udemy, and blogs
- Huge community of designers sharing templates and presets
- Easy to find solutions to problems